

Keyu Chen

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 CurryChen77 |  Google Scholar

Beijing, China

RESEARCH INTERESTS

My research centers on **traffic simulation, motion planning, and closed-loop RL fine-tuning** for autonomous driving, with a particular emphasis on creating **realistic, interactive, and controllable traffic scenarios**. I am passionate about developing reliable closed-loop simulation frameworks to accelerate the safe and robust advancement of autonomous driving.

EDUCATION

- **Tsinghua University (THU)** September 2023 - July 2028(Expected)
Beijing, China
Ph.D. student at the School of Vehicle and Mobility.
 - Advisor: [Prof. Sifa Zheng](#).
- **Nanjing University of Aeronautics and Astronautics (NUAA)** September 2019 - July 2023
Nanjing, China
B.Eng. in Vehicle Engineering.
 - GPA: 4.1/5 (Top 2%)

FIRST AUTHOR PUBLICATIONS

- [arXiv 2026] CRAFT: Counterfactual-to-Interactive Reinforcement Fine-Tuning for Driving Policies.
Keyu Chen, Nanfei Ye, Yida Wang, Wenchao Sun, Danqi Zhao, Hao Cheng, Sifa Zheng.
[Paper](#), [Project Page](#), [Code](#)
CRAFT improves driving policies by combining dense counterfactual proxy supervision with residual correction from true closed-loop interaction.
- [ICRA 2026] ForSim: Stepwise Forward Simulation for Traffic Policy Fine-Tuning.
Keyu Chen, Wenchao Sun, Hao Cheng, Zheng Fu, Sifa Zheng.
[Paper](#), [Project Page](#), [Code](#)
ForSim achieves high-fidelity traffic simulation by combining stepwise closed-loop trajectory propagation with physical dynamics and interaction-aware agent updates.
- [arXiv 2025] RIFT: Group-Relative RL Fine-Tuning for Realistic and Controllable Traffic Simulation.
Keyu Chen, Wenchao Sun, Hao Cheng, Sifa Zheng.
[Paper](#), [Project Page](#), [Code](#),  Stars 130+
RIFT achieves realistic and controllable traffic simulation by combining IL pre-training in a data-driven simulator for realism with RL fine-tuning in a physics-based simulator for controllability.
- [CoRL 2024] FREA: Feasibility-Guided Generation of Safety-Critical Scenarios with Reasonable Adversariality.
Keyu Chen, Yuheng Lei, Hao Cheng, Haoran Wu, Wenchao Sun, Sifa Zheng. (Oral 4.3%)
[Paper](#), [Project Page](#), [Code](#),  Stars 60+
FREA incorporates feasibility as guidance to generate adversarial yet AV-feasible, safety-critical scenarios for autonomous driving.
- [KBS 2023] IGT: Illumination-guided RGB-T object detection with transformers.
Keyu Chen, Jinqiang Liu, Han Zhang. (Bachelor's thesis)
[Paper](#)
IGT uses illumination intensity to guide the fusion process of multi-modality features, enabling the comprehensive utilization of cross-modality complementary information in object detection.

CO-AUTHOR PUBLICATIONS

- [arXiv 2026] SparseDriveV2: Scoring is All You Need for End-to-End Autonomous Driving.
Wenchao Sun, Xuewu Lin, **Keyu Chen**, Zixiang Pei, Xiang Li, Yining Shi, Sifa Zheng.
[Paper](#), [Code](#)
- [arXiv 2025] DriveCamSim: Generalizable Camera Simulation via Explicit Camera Modeling for Autonomous Driving.
Wenchao Sun, Xuewu Lin, **Keyu Chen**, Zixiang Pei, Yining Shi, Chuang Zhang, Sifa Zheng.
[Paper](#), [Code](#)
- [TRC 2025] Emergency Index (EI): A two-dimensional surrogate safety measure considering vehicles' interaction depth.
Hao Cheng, Yanbo Jiang, Hailun Zhang, **Keyu Chen**, Heye Huang, Shaobing Xu, Jianqiang Wang, Sifa Zheng.
[Paper](#), [Code](#)

HONORS AND AWARDS

- Outstanding Graduate (Top 1%), NUAA *June 2023*
- Outstanding Bachelor Thesis Award (Top 5%), NUAA *June 2023*
- National Scholarship (Top 1%), NUAA *December 2021*